Keshav Sharma

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EDUCATION

Vellore Institute of Technology — **Chennai, Tamil Nadu**

Bachelor of Technology in Computer Science (Data Science Specialisation) — *Graduation: May 2027*

SKILL SUMMARY

Languages & Core: Python, SQL, Java, C/C++, R
Data Science & Statistics: Machine Learning, Deep Learning (PyTorch), Statistics & Probability, Scikit-learn, Pandas, NumPy, Feature Engineering, Model Evaluation, Time-Series Analysis, Anomaly Detection, Probabilistic Modeling (BG/NBD, Gamma-Gamma)
Libraries & Tools: Pandas, NumPy, Scikit-learn, Matplotlib, Power BI, Tableau
Generative AI: LLMs, Prompt Engineering, Retrieval-Augmented Generation (RAG), Google Gemini API
Engineering Practices: Git/GitHub, PyTest, Test-Driven Development (TDD), Jupyter Notebook, Reproducible Experimentation

EXPERIENCE

VOIS – Vodafone Idea Foundation* |LINK** — ***Aug – Sep, 2025

- Collected and analyzed 2 real-world datasets (8,500+ records) using LLM-based natural-language querying, extracting 15 structured business insights.
- Iteratively validated AI-generated outputs for statistical consistency and accuracy, ensuring findings were reliable before use.
- Designed a repeatable LLM-assisted analytics workflow combining data preprocessing, natural-language querying, insight extraction, and validation to convert raw datasets into actionable business findings.

PROJECTS

***Customer Lifetime Value Engine* |LINK**

- Processed 1M+ historical transaction records and implemented BG/NBD + Gamma-Gamma probabilistic models to forecast customer purchase frequency, monetary value, and lifetime value over a 12-month horizon.
- Segmented customers into 5 CLV-based groups, identifying the top 20% of customers contributing ~65% of projected revenue, enabling targeted analysis of high-value customer cohorts.
- Built a Power BI dashboard tracking 8+ KPIs across 12 customer cohorts, visualizing CLV distribution, repeat-purchase behavior, retention patterns, and projected revenue.

***Real-Time Stock Market Anomaly Detector* |LINK**

- Engineered 15+ time-series features from 500K+ NSE market observations and trained Isolation Forest and LSTM Autoencoder models to detect anomalous price and volume movements.
- Applied FinBERT sentiment analysis to 25K+ financial news records and integrated sentiment scores with market features, combining structured market data and unstructured financial text into a unified analytical pipeline.
- Benchmarked 2 anomaly-detection models using precision, recall, F1-score, and inference time, with the best model targeting ~86% precision, ~81% recall, and ~83% F1-score on the evaluation dataset.

***Docwise* |LINK**

- Built a RAG-based document Q&A system using Google Gemini, NLP, TF-IDF retrieval, and intelligent chunking, achieving 95%+ query accuracy on PDFs exceeding 200 pages.
- Designed an end-to-end document processing and retrieval pipeline for text extraction, chunking, vectorization, context retrieval, and LLM-based response generation.
- Optimized retrieval quality through prompt engineering, relevance tuning, and hallucination-aware validation, improving response reliability and contextual accuracy.

***AI-Powered Expense Receipt Scanner & Analytics Platform* |LINK**

- Built an OCR-based data extraction pipeline using EasyOCR, OpenCV, and NLP to transform unstructured receipts into structured transaction data, improving processing efficiency by 40%.
- Developed data preprocessing and expense categorization workflows to clean, validate, and organize extracted transaction data for downstream analysis.
- Built analytics and reporting features for spend trends and budget tracking, converting transaction data into actionable financial insights for end users.

Certifications

- Oracle Cloud Infrastructure 2025 Certified Generative AI Professional |**LINK** 2026
- Prepare Data for ML APIs on Google Cloud Skill Badge |**LINK** 2026